

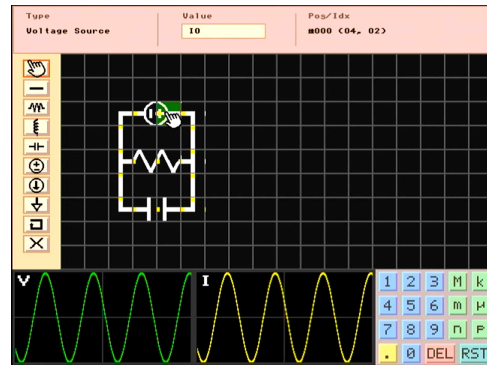
Sheet A: Component / Capability description part

Team: S1_G9

An Interactive Circuit Design & Visualization System

86 Words

MetaCircuit is an interactive FPGA-based circuit design system that enables users to construct, edit, and visualize circuits in real time. Through a **mouse-driven VGA** interface, users can place and modify components such as resistors, capacitors, inductors, and independent voltage/current sources on a digital canvas. Users can also modify component parameters on the property editor through a virtual keyboard. The system also provides integrated supporting functions, including matrix displays (SW[0]), component storage visualization (SW[1]), waveform plotting, and an **OLED**-based mini calculator for simple arithmetic.



System Overview

Student A: Chen Shimin (A0311929B)

Circuit Canvas & Component Storage

101 Words

Circuit Canvas: Stores circuit cells in memory, mapping them onto grids, and displaying them reliably on a 640x480 **VGA** output.

Component Store: Manages a 40-bit parameter representation for every component, supporting real-time visual updates on the display (SW[1]). This data structure encapsulates critical attributes, such as component type, coordinates, component index, and orientation.

VGA Protocol: Static circuit renderer for the Basys3 **VGA** pipeline. **VGA** module allows FPGA to use **VGA** output and display the whole UI design on a bigger screen. It is important to set the timing, HSync, and Vsync such that the screen can output the proper image.



Circuit Canvas



Component Store

Student B: HSU YUCHEN (A0312064U)

Waveform Visualization & Dynamic Current Display

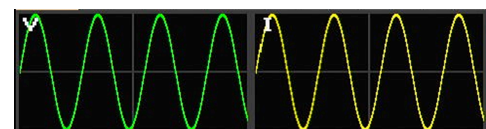
114 Words

Real-Time Waveform Plot: The real-time waveform plot is meant for users to understand the behavior of the circuit drawn on the circuit Canvas. The waveform takes the value from LUT and displays it on the plot.

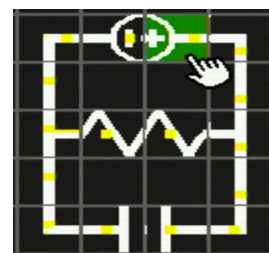
Dynamic Current Display: The dynamic current display is designed to show users the visual representation of the direction of the current flow. The display will read the component type from the RAM and update the **VGA** output.

Ping Pong Buffer: Ping Pong Buffer acts as an intermediate step before **VGA** rendering, where the **VGA** output will switch between two buffers, ensuring the output completion.

Panning: Allows the grid to move and show the rest of the circuit canvas.



Waveform Plot



Dynamic Current Display

Student C: Xie Ziyi (A0311479B)	Interactive UI & System Control	101 Words																																																																															
Toolbar & Draw: Toolbar with mode-based interaction; each tool (e.g., wire, resistor) is mapped to a distinct drawing state (e.g., component placement, rotation, and deletion). Cursor Design: Customizes default cursor into a hand-style icon with hover and pressed states. Keyboard & Input: Supports numeric, decimal, and unit input via a virtual keyboard for parameter configuration. Property Panel Interaction: Dynamically updates property panel based on mouse state. Selecting a component displays its corresponding parameters; virtual keyboard interaction enables value input; otherwise, panel shows “Click component”. LED Display: Displays number of components placed on canvas, with the same number of LEDs illuminated accordingly.	 Drag  Wire(s)  Resistor  Inductor  Capacitor  Voltage Source  Current Source  Ground  Rotate  Delete  Hover  Pressed Value 100p Input Display 123Mk 456mW 789nF .0DEL RST Keyboard  Canvas Draw																																																																																
Student D: Wang Zhirui (A0303291N)	Matrix Display & User Interaction	92 Words																																																																															
Matrix Display: Displays two matrices stacked vertically above canvas when needed. Each cell renders a 2 d.p. floating number in the range -999.99 to 999.99. Dynamic Text display: Supports runtime-modifiable text display via a flattened 256-bit bus, with customized magnification functions and Q8.8 fixed-point numbers to ASCII conversion. Font Rom: 8x8 pixel character font (ASCII 32-126) and Compact 5x7 pixel digit font. Calculator: Complete OLED calculator with a navigable keyboard interface and Overflow/underflow error detection. Implemented using a 10-stage ripple-carry adder/subtractor built from cascaded 1-bit full adders.	Type Voltage Source Value 10 Pos/Idx #000 C03, 023 Component Property Panel  OLED Calculator <table><tr><td>-128.00</td><td>2.25</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>2.25</td><td>-20.25</td><td>3.50</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>0.00</td><td>3.50</td><td>30.10</td><td>4.75</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>0.00</td><td>0.00</td><td>4.75</td><td>40.25</td><td>5.89</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>0.00</td><td>0.00</td><td>0.00</td><td>5.89</td><td>-51.40</td><td>6.69</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td><td>6.69</td><td>60.80</td><td>7.39</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td><td>6.69</td><td>7.39</td><td>71.10</td><td>8.10</td><td>8.10</td></tr><tr><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td><td>8.10</td><td>8.10</td><td>99.98</td><td>99.98</td></tr></table> Matrix Display	-128.00	2.25	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.25	-20.25	3.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.50	30.10	4.75	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.75	40.25	5.89	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.89	-51.40	6.69	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6.69	60.80	7.39	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6.69	7.39	71.10	8.10	8.10	0.00	0.00	0.00	0.00	0.00	0.00	8.10	8.10	99.98	99.98
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